



Predicting Pokémon Power: STOR 664 Final Project Introduction:

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Our Data: Preliminary EDA and Hypothesis



We have elected to use a dataset containing stats and other key information for characters from the popular Japanese video game series Pokémon.

Dataset Snapshot – Key Variables



Height



Weight



Attack



Defense



Special Attack



Special Defense



Base Experience



Health Points (HP)

Our Hypotheses to Test

- Since HP is one of the most important stats in determining the power of a Pokémon, we decided to focus on the relationship and influence of other key stats on HP.
- After our preliminary EDA phase, we determined the following hypotheses as potentially interesting to examine:

$$H_0^{(1)}: \beta_{\text{speed}} = \beta_{\text{attack}} = 0$$

$$H_0^{(2)}: \beta_{\text{dual_type_x_defense}} = 0$$

$$H_0^{(3)}: \beta_{\text{base_experience}} = 0$$

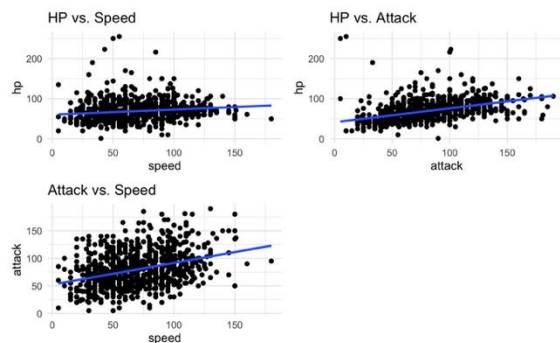
Our key question: What stats are most predictive for total HP?

Preliminary EDA: Influential Plots

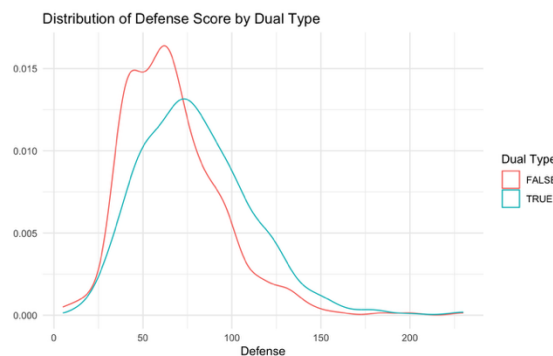


Below are a selection of interesting plots that influenced our selection of hypotheses.

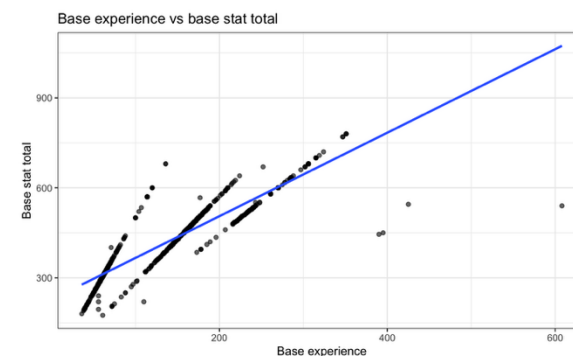
Bivariate Correlations: HP vs Speed vs Attack



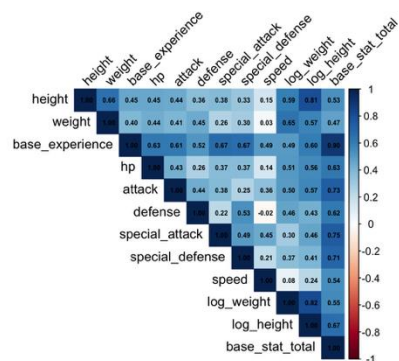
Defense vs Dual Type Distribution



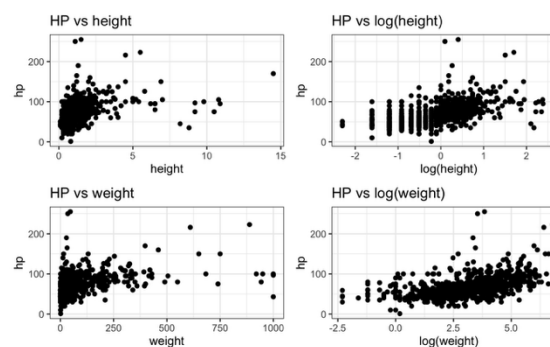
Base Experience vs Total Base Stats



Correlation Matrix



HP vs Size Stats



Our Plan: Further Analysis Steps



The group will perform predictive analysis on the data and conduct hypothesis tests to determine the answers to our question.

Step 1: Iterative Process to Determine Proper Linear Model

The team will perform an iterative, back-and-forth model selection procedure on the data to determine which variables/interactions are significant in a linear model predicting HP.

Step 2: Hypothesis Testing

Using methods discussed in class (T- and F- testing), the team will perform hypothesis testing on the fitted and optimized models to reject/fail-to-reject the 3 hypotheses outlined earlier.

Step 3: Conclusions

Based on the outcomes of the hypothesis testing, the group will draw conclusions and summarize them in the final project writeup.

A large, dense crowd of various Pokémon characters is gathered together in a celebratory pose. The characters are diverse in species and color, including Pikachu, Eevee, and many others. They are all looking towards the center of the image where the text is located. The background is a dark, starry space.

Thank you!

Any Questions?